K. M. Mynurdoza

Department of Mechanical Engineering · Bangladesh University of Engineering and Technology (BUET)

Robotics & Automation · Multiphysics Simulation (FEA/CFD) · Computational Mechanics

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Executive Profile

Mechanical Engineering undergraduate at Bangladesh University of Engineering and Technology (BUET) with research-driven expertise in modeling, simulating, and automating complex physical systems. Proficient in multiphysics finite element analysis (FEA), conjugate heat transfer, precision mechanism synthesis, and embedded real-time motion control architectures. Demonstrated academic rigor through national and international science Olympiad accolades, hands-on robotic instrumentation, and open-source computational engineering software. Actively seeking advanced research collaborations and graduate-level engineering opportunities.

Education

Bangladesh University of Engineering and Technology (BUET) — **Dhaka, Bangladesh**
Bachelor of Science in Mechanical Engineering — *Nov 2023 – Expected Jul 2028*

- **Academic Standing:** Level-3, Term-II · Completed **80.00 credit hours** across 4 completed academic terms.
- **Cumulative Grade Point Average: 3.70 / 4.00** · Recent Term GPA: **3.83 / 4.00** (Level-2, Term-II).
- **Curricular Concentration:** Continuum mechanics, thermodynamics, fluid dynamics, mechatronics, and numerical simulation.

Sylhet Cadet College — **Sylhet, Bangladesh**
Higher Secondary Certificate (HSC) – Science — *Passing Year: 2022*

- **Result: GPA 5.00 / 5.00 (Golden A+)** · Awarded Government Talentpool Scholarship.
- **Distinctions: First Boy** of the college cohort.

Sylhet Cadet College — **Sylhet, Bangladesh**
Secondary School Certificate (SSC) – Science — *Passing Year: 2020*

- **Result: GPA 5.00 / 5.00 (Golden A+)** · Awarded Government Talentpool Scholarship.
- **Distinctions:** Ranked **7th** in the entire Sylhet Education Board merit list.

Academic Disciplines & Experimental Foundations

Comprehensive theoretical coursework reinforced by hands-on laboratory experimentation at BUET:

- **Thermal & Fluid Sciences:** Thermodynamics, Conduction & Convective Heat Transfer, Fluid Mechanics (with dedicated experimental fluid mechanics and thermal engineering laboratories).
- **Solid Mechanics & Machine Design:** Mechanics of Materials, Engineering Mechanics (Statics & Dynamics), Machine Elements Design (materials testing, stress-strain characterization, and failure analysis).
- **Instrumentation, Controls & Systems:** Measurement & Instrumentation, System Dynamics & Feedback Controls, Sensor Interfacing (hands-on transducer calibration and data acquisition laboratories).
- **Computational & Manufacturing Methods:** Numerical Methods, Computer Programming in C/C++, CAD/CAM, Machine Shop Practice, Foundry & Welding Engineering.

Research & Engineering Projects

Finite Element Thermal Analysis of Oil-Cooled DC-Link Film Capacitors *Ongoing Research*

Focus: Multiphysics Simulation · Conjugate Heat Transfer · Power Electronic Packaging

- Investigating conjugate heat transfer and thermal-fluid behavior in dielectric oil-submerged DC-link metallized polypropylene film capacitors utilized in heavy-duty traction converters.
- Constructing 3D finite element numerical models to simulate internal electro-thermal dissipation, core hotspot genesis, and convective dielectric fluid circulation.
- Evaluating internal cooling channel geometries, thermal boundary layers, and forced circulation flow paths to mitigate dielectric breakdown risks and enhance operational lifespan.

Thermomechanical Simulation of Power Module Solder Joints *FEA / Reliability Modeling*

Technologies: COMSOL Multiphysics, Thermal-Structure Interaction, Anand Viscoplasticity, Morrow Fatigue

- Replicated failure mechanisms and stress analysis of a Single-Sided Cooling (SSC) power module (Semikron packaging) using a 2D half-symmetry coupled thermal-structural FEA model.
- Implemented the nonlinear **Anand Viscoplasticity Model** for 60Sn40Pb solder to capture strain-rate- and temperature-dependent inelastic deformation under cyclic thermal loading ($\Delta T_j \approx 120°C$).
- Executed segregated solvers with strict time stepping and targeted solder corner mesh refinement, evaluating peak Von Mises stresses (150–160 MPa).
- Quantified plastic dissipation energy density per thermal cycle (ΔW_{sc}) and integrated with the **Morrow fatigue model** to predict solder joint thermal cycling lifetime (N_{sf}).

"Beyond Horizon" – Automated Smart Alt-Azimuth Robotic Mount *2024 – 2026*

Technologies: SolidWorks, COMSOL Multiphysics, ESP32, C++, Stepper Motion Control **GitHub Repository**

- Designed, optimized, and fabricated a dual-axis motorized robotic mount for a custom refractor telescope, achieving automated target acquisition and real-time celestial tracking.
- Synthesized high-ratio precision reduction gear trains in SolidWorks, balancing payload torque capacity, minimal mechanical backlash, and vibration dampening during long-exposure tracking.
- Implemented embedded C++ firmware executing real-time coordinate transformations from equatorial coordinates (Right Ascension / Declination) to local topocentric horizon coordinates (Altitude / Azimuth).
- Performed finite element structural deflection and thermal stress analysis on the D80F900 doublet optical lens assembly in **COMSOL Multiphysics** to preserve optical axis collinearity under gravity sag and thermal gradients.

Beam Deflection Solver via Macaulay's Singularity Method *Computational Mechanics*

Technologies: Python, NumPy, Matplotlib, Boundary Value Formulations **GitHub Repository**

- Developed a Python-based structural analysis engine using Macaulay's bracketed singularity functions to determine elastic deflection profiles under complex combinations of point loads and moments.
- Formulated a unified boundary-value integration scheme across piecewise loading regimes, eliminating manual multi-segment boundary matching and direct constant resolution.
- Automated graphical rendering pipelines for continuous shear force diagrams (SFD), bending moment diagrams (BMD), and elastic slope/deflection curves.

Rough Khata – STEM Typography & Live LaTeX Note-Taking Engine *Software Architecture*

Technologies: TypeScript, React 19, Vite, Electron, KaTeX, HTML5 Canvas **GitHub Repository**

- Architected a high-performance cross-platform desktop application designed for mathematical derivation, STEM document annotation, and live digital instruction.
- Engineered an instant LaTeX typesetting engine utilizing KaTeX coupled with low-latency stylus and canvas rendering to handle intricate multi-line formulas seamlessly.
- Implemented smooth PDF annotation pipelines, infinite canvas panning, and vectorized exports for publication-quality instructional notes.

Numerical Simulation of Non-Linear Multi-Degree Dynamic Systems *Independent Project*

Technologies: Python, SciPy, C++, ODE Integration, Phase-Space Analysis

- Implemented numerical integration routines (4th-order Runge-Kutta and adaptive step-size solvers) to model forced non-linear oscillators and coupled multi-body mechanical systems.
- Evaluated dynamic response regimes, non-linear damping dissipation, bifurcations, and phase-space trajectories across transient and steady-state excitations.

Honors, Awards & Olympiad Distinctions

- **National Round Winner** · **Bangladesh Physics Olympiad (BdPhO)**, 2022
  Selected for the intensive National Coaching Camp in the International Physics Olympiad (IPhO) selection pipeline.
- **Global Award of Merit** · **6th International Blockchain Olympiad (IBCOL)**, Amsterdam, Netherlands, 2023
  Represented Bangladesh globally (Team GreyDevs); presented an enterprise decentralized engineering architecture.
- **Divisional Winner & Selection Winner** · **Bangladesh Mathematical Olympiad (BdMO)**, 2022 & 2023
  Achieved top honors across multiple national rounds governed by the Bangladesh Math Olympiad Committee.
- **Round 1 Qualifier** · **Google Code Jam**, 2022 | **Accomplishment Certificate** · **Google Kick Start**, 2021
  Successfully solved advanced algorithmic optimization and competitive programming problems in Google's global rounds.
- **World Rank 18th** · **International Mental Math Challenge**, 2022 · Global high-speed arithmetic deduction.
- **Certificate of Achievement** · **11th Bangladesh Chemistry Olympiad**, 2020 · National achievement in physical/organic chemistry.
- **Certificate of Achievement** · **Bangladesh Biology Olympiad (BDBO)**, 2020 · Divisional achievement awardee & national qualifier.
- **Government Talentpool Merit Scholarships** · Ministry of Education, Bangladesh (Both HSC 2022 and SSC 2020).

Technical Skills Matrix

- **CAE & Multiphysics Simulation:** COMSOL Multiphysics (Structural Mechanics, Heat Transfer, Fluid Flow Coupling), ANSYS (Fluent, Mechanical), Finite Element Analysis (FEA), Computational Fluid Dynamics (CFD).
- **CAD & Mechanical Prototyping:** SolidWorks (Parametric 3D Solid Modeling, Assemblies, Motion Analysis), AutoCAD, GD&T Principles, Rapid Prototyping (FDM 3D Printing, Laser Cutting).
- **Robotics, Embedded Systems & Control:** ROS / ROS 2, Embedded C/C++, Microcontrollers (ESP32, STM32, Arduino), Motion Control (Stepper/Servo Actuation), Sensor Interfacing (IMU, Encoders, Thermocouples), Closed-Loop PID Control.
- **Programming Languages:** Python (NumPy, SciPy, Matplotlib), C, C++, MATLAB, TypeScript, SQL, HTML/CSS.
- **Development & Scientific Tools:** Git/GitHub, Linux/UNIX, LaTeX Typography, VS Code, Electron, Vite.
- **Laboratory & Experimental Methods:** Tensile & Compression Testing, Rockwell/Brinell Hardness, Impact Testing (Charpy/Izod), Sensor Calibration, Fluid Mechanics Test Benches.

Leadership, Professional Affiliations & Extracurricular Activities

- **Student Affiliate Member** · **Institution of Mechanical Engineers (IMechE), UK**
  Active participant in international technical webinars, mechanical engineering symposiums, and BUET IMechE student chapter initiatives.
- **Official Campus Ambassador** · **Chess.com** (Academic Year 2026 – 2027) — Appointed university ambassador representing BUET; organizing campus chess tournaments, collegiate workshops, and student outreach.
- **Treasurer** · **BUET Chess Club** (Jul 2026 – Present) — Directing club financial management, annual budget allocations, sponsorship acquisition, and tournament operations for university-wide and national inter-collegiate championships.
- **FIDE-Rated Competitive Chess Player** · **International Chess Federation (FIDE)** — Officially rated tournament chess player; representing institutional teams in national rating events. Champion in Inter-House Chess at Sylhet Cadet College.
- **Editorial & Writing Contributions** · **Sylhet Cadet College** — Authored featured analytical and scientific articles for the college annual journal; recognized in inter-house literary competitions.

Languages & Personal Interests

- **Languages:** Bengali (Native Speaker) · English (Full Professional Working Proficiency).
- **Interests:** Computational physics, astronomy & observational astrophysics, chess strategy, hiking, exploration & travel.